

PROJECT SPECIFICATION: LIBRA-LOCAL (CORE BLUEPRINT)

Target Environment: iOS (Via Tauri Cross-Compilation v2)
Classification: Premium, Local-First, Low-Resource Open-Source Utility

1. System Architecture & Guardrails

- Complete Network Isolation:** The application must function at 100% capacity in total offline environments. No external analytics, font CDNs, or remote verification hooks are permitted.
- Silicon Longevity Optimization:** Maximize UI main-thread responsiveness for legacy or hand-me-down mobile chipsets. The engine must actively reject heavy client-side JavaScript execution loops or bloated rendering wrappers.
- The Content Hashing Protocol:** Because archived/community files are frequently moved, renamed, or restructured by users within the native iOS Files App, files must never be cataloged by volatile path strings. Every document receives an unalterable SHA-256 hash calculated asynchronously from its raw binary content upon initial discovery. All annotations, progress indicators, bookmarks, and custom asset pairings lock exclusively to this content hash.

2. Codebase Architecture Directory Structure

```
libra-local-project/  
■■■■ src-tauri/ # Systems Layer (Rust Host Engine)  
■ ■■■■ src/  
■ ■ ■■■■ main.rs # Application Lifecycle & Command Router  
■ ■ ■■■■ commands/ # Thread-Isolated Core Capabilities  
■ ■ ■ ■■■■ library.rs # Async Directory Traverser & Tree Scanner  
■ ■ ■ ■■■■ reader.rs # Progress Tracker & Annotation Manager  
■ ■ ■ ■■■■ crypto.rs # Non-blocking SHA-256 Calculation Utility  
■ ■ ■■■■ storage/ # Transactional Atomic JSON File Engine  
■ ■■■■ Cargo.toml  
■■■■ src/ # Interface Layer (Webview UI)  
■ ■■■■ core/ # Pure Abstraction Services  
■ ■ ■■■■ navigation/ # Centralized SPA Router State Machine  
■ ■ ■■■■ zoom/ # Scale-Then-Render Debounce Controller  
■ ■ ■■■■ events/ # Native Selection Capture Hub  
■ ■■■■ components/ # Rigid UI Primitives (Token-Compliant Only)  
■ ■ ■■■■ Library/ # Folder System Tree & Metadata Display  
■ ■ ■■■■ Reader/ # Viewport Container & Invisible Text Layer  
■ ■ ■■■■ Common/ # Global Navigation Bar & Action Dropdowns  
■ ■■■■ styles/  
■ ■ ■■■■ tokens.css # Strict System Design Tokens Colors/Spacing  
■ ■ ■■■■ main.css # Global Framework Resets  
■ ■■■■ index.html  
■ ■■■■ main.ts
```

3. Core Tool Modules & Interaction Framework

- 3.1 Library View: Folder Mirroring & Metadata Anchoring**
The Rust systems layer monitors an un-sandboxed directory in the public file structure, automatically reflecting modifications in real time without altering the source files.
- 3.2 Book Reader View: Engine & State Management**
Utilizes a decoupled PDF.js integration inside the Tauri frontend view frame for high-speed, local processing. Handles

Table of Contents via outline extraction hooks and pushes page-state metrics dynamically to backend actors.

3.3 The Scale-Then-Render Zoom Engine

To completely sidestep text clipping or frame-rate stuttering on low-spec mobile chipsets, zooming operates via a two-stage performance pipeline: immediate hardware-accelerated CSS scales at 60fps, followed by a 300ms debounced vector redraw pass.

3.4 Hardware-Accelerated Reading Themes

To preserve battery life and eliminate heavy re-rendering tasks during theme updates, the interface implements system-wide reading filters directly via GPU-accelerated CSS layers:

```
.pdf-viewport-frame[data-theme='night'] .pdf-page-canvas-render { filter: invert(1) hue-rotate(180deg); }
```

3.5 Floating Selection & Sliding Annotation Panel

Intercepts native events on the transparent, overlaid HTML text layer. Tapping 'Annotate' triggers the global SPA router to slide open a dedicated input canvas from the left margin, caching data against the core text signature string.

4. Universal Design System Token Matrix

Token Parameter	Enforced Value	Context
--spacing-tight	0.5rem (8px)	Inner element margins, label buffers
--spacing-element	0.75rem (12px)	Toolbar configurations, button gutters
--spacing-card	1.0rem (16px)	Library thumbnails, annotation text blocks
--spacing-container	1.5rem (24px)	Master window outer boundaries
--radius-interactive	12px	Clickable button elements, select fields
--radius-container	20px	Folder tiles, sliding contextual sheets

5. Master AI System Prompts & Guardrails (SKILL.md)

1. UNIVERSAL DESIGN ENFORCEMENT
- You are explicitly forbidden from using raw pixel integers or inline styles for layout geometry.

- Every padding, margin, gap, and edge offset must resolve directly through a --spacing-* CSS token.
2. LOGICAL ISOLATION (ZERO DECOUPLING LOOPS)
- Do not write unique inline navigation events or create custom, localized back buttons.

- All visual transitions must interface directly with the centralized 'UniversalRouter' state stack.
3. DATA PROTECTION & DISK TRANSACTIONS
- Visual templates are strictly barred from initializing direct filesystem actions.

- All file tree crawling, content evaluations, hashing metrics, and data writes must process asynchronously through native Rust Tauri command handlers.